

TEJ GANGUPANTULA

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EDUCATION

University of California, Santa Barbara

September 2023 - December 2026

B.S. Statistics and Data Science

Santa Barbara, CA

- **GPA:** 3.6
- **Relevant Courses:** Statistical Machine Learning, Regression Analysis, Time Series Analysis, Survival Analysis, Stochastic Processes, Database Management Systems, Data Analysis and Inference, Data Science with R/SQL, Probability and Statistics, Calculus I-III, Linear Algebra, Differential Equations

SKILLS

AI/Agents: LangGraph, LangChain, MCP (TS SDK), Pydantic, Zod, Gemini/Vertex AI, Anthropic API, OpenAI API

Languages: Python, R, TypeScript/Node, JavaScript, SQL, Java

ML/Data: Scikit-learn, TensorFlow, spaCy/medspaCy, pandas, NumPy, Matplotlib, Seaborn

Infra/Tools: Google Cloud Platform (Cloud Run), Docker, FastAPI, PostgreSQL, Git, Streamlit, Jupyter

PROFESSIONAL EXPERIENCE

AI Engineer Intern

June 2025 - Present

Agentman (Three Internship Terms: Summer 2025, Spring & Summer 2026)

San Francisco, CA

- Architected and shipped production MCP (Model Context Protocol) servers in TypeScript/Node, exposing 30+ Zod-validated tools that let LLM agents operate Gmail (search, compose, send, label) and a Gemini-backed image-generation service with OAuth-brokered auth, tenant-scoped file storage, and a stateless architecture that scales horizontally on GCP Cloud Run.
- Built agentic AI workflows for healthcare operations on GCP (HIPAA-compliant), configuring conversational and flow-style agents backed by LangGraph/Chain orchestration to automate clinical administrative tasks end-to-end.
- Wired EHR data sources, third-party healthcare APIs, and vendor-hosted MCP servers into the agents' tool layer via HTTP and MCP tool integrations, with Pydantic schema validation on every tool call for reliability, reducing eligibility verification from 50 → 7 hrs/month and per-check cost from ~\$6.72 to \$0.50.

Machine Learning Intern

July 2024 - September 2024

FusionCare

Davis, CA

- Built a Python-based clinical text processing pipeline to ingest, clean, and structure de-identified patient notes, applying NLP-based PII masking (spaCy/medspaCy) to safely prepare data for downstream analysis.
- Integrated OpenAI GPT models via LangChain with structured prompting and output validation to convert unstructured clinical notes into JSON-formatted visit summaries, prototyping outputs using Streamlit, improving care plan evaluations and saving clinicians 4-5 hours weekly.

PROJECTS

Prior Authorization Voice Agent | *Python, LangGraph, LangChain, FastAPI, PostgreSQL*

May 2026 - June 2026

- Built a real-time voice agent (STT → LangGraph state machine → TTS) that answers prior-authorization status questions over a browser interface, routing callers across four normalized outcomes via LangGraph's conditional edges rather than an open-ended ReAct loop.
- Integrated the ElevenLabs API for real-time speech-to-text and text-to-speech, and designed three consent-gated tools (PA lookup, appeal ticket, escalation ticket) backed by a PostgresSaver checkpointer for durable per-call state that survives process restarts.
- Authored an architecture-first design that pairs every demo shortcut with its production successor, and implemented structured error handling with per-tool routing and a dedicated escalation path for total failures.

NBA Finals Outcome Prediction Analysis | *Python, Scikit-learn, NumPy, pandas*

April 2025 - June 2025

- Developed a Random Forest classification model in Python to simulate and predict 2025 NBA Finals outcomes, achieving 62% test accuracy using team-level regular season and playoff metrics.
- Engineered a Monte Carlo simulation to estimate championship probabilities over a best-of-seven series, predicting a 75.5% chance of the Oklahoma City Thunder winning against the Indiana Pacers (24.5%).
- Cleaned, processed, and analyzed historical NBA data with pandas, numpy, and scikit-learn, enabling interpretable model outputs and supporting data-driven sports analytics.